Aman Behera

+91-9587987718 | aman.behera.systems@gmail.com | beingamanforever.github.io | github.com/beingamanforever

Education

Indian Institute of Technology, Roorkee Oct 2022 – May 2026
Bachelors in Technology — Chemical Engineering *CGPA: 8.124 / 10*

Experience

ML Systems Intern | Infoedge India Ltd May 2025 – Jul 2025

- Worked on Generative Engine Optimization (GEO) to improve content visibility in LLM-powered search by fine-tuning Qwen3-8B-Instruct using SFT and DPO for higher fluency, credibility, and citation quality.
- Designed objective metrics (position-adjusted word count, sentence position) and GEval-GPT based subjective evaluation. Built workflow with rewriting agents & evaluation modules for testing strategies.

Research Intern | XLRI Xavier Institute of Management Dec 2023 – Jan 2024

- Built a user-space memory system simulator modeling physical allocators, virtual memory with paging, and multi-level CPU caches to analyze the end-to-end path from virtual address translation to cache hit/miss behavior
- Implemented LRU-based page replacement and set-associative L1/L2 cache models, collecting fragmentation, page fault, and cache performance metrics to study memory hierarchy trade-offs.

Projects

ZeroShred: P2P File Transfer System Sep 2025 – Jan 2026

- Built a high-performance P2P file transfer system in C++ using zero-copy I/O primitives to stream data directly from disk to network, achieving up to 780 MB/s throughput while minimizing user–kernel data movement.
- Performed SHA-256 integrity verification in a parallel thread, decoupled from the data transfer path.

CSV Parser Engine in C++ Oct 2025 – Dec 2025

- Implemented a high-throughput CSV parser in C++ using mmap-based file ingestion, allocation minimized parsing, and delimiter detection via bitwise operations to maximize sequential memory throughput.
- Designed cache-friendly structure-of-arrays (SoA) memory layouts and compared parsing throughput against standard line-by-line CSV reading approaches.

Binary Protocol Encoder/Decoder in C++ Jul 2025 – Aug 2025

- Designed a fixed-layout binary wire protocol and implemented explicit field-by-field encoding/decoding with endian-safe serialization to ensure portability and predictable performance.
- Used offset-based packing to avoid struct memcpy and dynamic allocation, reducing message handling overhead.

Academic Achievements & Competitions

- Secured **AIR 4759** in JEE Advanced 2022 among 1.5 lakh candidates
- Secured **Global Rank 1075** in Codeforces Round 994 (Div. 2)
- **Silver Medal**, Inter IIT Tech Meet 2025, representing IIT Roorkee
- **Winner**, AIQuest Hackathon 2025 conducted by **HiLabs Inc**
- Ranked **67th** (Task 5) and **24th** (Task 2) in **highload.fun** performance engineering challenges

Olympiad Experience

- Awarded with **NTSE Scholarship**, 2020 by NCERT
- **Kishore Vaigyanik Protsahan Yojana (KVPY)**, 2021, **AIR 122** in SA Stream
- **Indian Olympiad Qualifier in Chemistry (IOQC)**, 2021, **Top 300** students nationwide
- **Indian National Chemistry Olympiad (INChO)**, 2020, **Top 300** students nationwide
- **Indian National Junior Science Olympiad (INJSO)**, 2019, **Top 100** students nationwide

Technical Skills

Programming Languages: C++, Python, SQL, Awk, Sed, Bash, Shell
Systems: Networking, OS Internals, Memory Management
ML / Data: PyTorch, Pandas, NumPy, FAISS, LLM fine-tuning, RAG

Positions of Responsibility

Project Head, ACM Student Chapter | IIT Roorkee Apr 2025 – Present

- Leading **50+** members; organizing talks, workshops, and projects in core CS and ML

Member, Vision And Language Group (VLG) | IIT Roorkee Jan 2024 – Present

- Participating in paper discussions, workshops, and projects focused on Deep learning, NLP, and CV

Competitive Programmer | Codeforces, LeetCode Aug 2024 – Present

- Codeforces max rating **1743** (Expert), handle: heWhoCooks; LeetCode rating **1704**, handle: beingamanforever